6.6 Integration by Parts

AP Calculus 6.6 · Student sheet

Name _____ Period ____ Date _____

I can…

- choose u and dv for integration by parts.
- use integration by parts, more than once if needed.
- evaluate a definite integral by parts.

Vocabulary · fill in the missing word

Integration by parts — ∫ u dv = uv − ∫ v du, the Product Rule run _____

Part 1 · Product Rule backward

Use the Product Rule to find an integral.

Step	Result
d/dx (x sin x)	
Solve for x cos x	
∫ x cos x dx	

Part 2 · Notes and examples (fill in each note, then work the example)

Integration by parts: ∫ u dv = _____ − ∫ v du.

Example 1

Find ∫ x cos x dx.

Choosing u: pick u so that du is _____ than u (logs and inverse trig first, then powers of x).

Example 2

Find ∫ ln x dx.

Repeat: if the new integral still has a product, use parts _____.

Example 3

Find $\int x^2 e^x \, dx$.

Definite integrals: ∫ from a to b of u dv = [uv] from a to b − ∫ from a to b of v du; evaluate uv at the _____.

Example 4

Evaluate $\int_0^1 xe^x\,dx$.

Part 3 · Partner practice

Partner 1

Find $\int xe^{2x}\,dx$.

Partner 2

(AP® Practice) $\int x \sin x\,dx =$

(A) $-x\cos x + \sin x + C$ (B) $x\cos x - \sin x + C$

(C) $-x\cos x - \sin x + C$ (D) $\left(\frac{x^2}{2}\right)\sin x + C$